

THE RAMANUJAN τ FUNCTION

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§1. Ramanujan

In a classic paper written in 1916, Ramanujan [26] introduced the τ function. It is defined by the power series expansion of an infinite product:

$$\sum_{n=1}^{\infty} \tau(n)q^n = q \prod_{n=1}^{\infty} (1 - q^n)^{24}.$$

Let $q = e^{2\pi iz}$ and set

$$\Delta(z) = \sum_{n=1}^{\infty} \tau(n)e^{2\pi inz}.$$

Then, $\Delta(z)$ is a modular form of weight 12 for the full modular group $SL_2(\mathbf{Z})$. In particular,

$$\Delta\left(\frac{az+b}{cz+d}\right) = (cz+d)^{12}\Delta(z) \quad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbf{Z}).$$

The purpose of this paper is to discuss various conjectures about the τ function enunciated by Ramanujan and others, and relate what is known and unknown about these conjectures at the present time.

Ramanujan was the first to foresee the arithmetical significance of $\tau(n)$, for he was the first to investigate the divisibility properties of these coefficients. From the work of Shimura, Serre, and Deligne, it is now known that the decomposition laws of primes in certain non-solvable extensions of $\mathbf{Q}$ are given by divisibility criteria for the τ function. The first indications of a general reciprocity law and non-abelian class field theory lie hidden in the divisibility properties of Fourier coefficients of cusp forms.

In [26], Ramanujan proposed three conjectures:

- (i) $\tau(n)$ is multiplicative: $\tau(nn') = \tau(n)\tau(n')$ whenever $(n, n') = 1$.
- (ii) if p is prime, $\tau(p^{n+1}) = \tau(p)\tau(p^n) - p^{11}\tau(p^{n-1})$, for $n > 1$, and
- (iii) $|\tau(p)| \leq 2p^{11/2}$.

Mordell [16] proved (i) and (ii) in 1917. Hecke [9] generalised his proof and unravelled the underlying theory. But (iii) defied many attempts. A succession of

analytic methods failed to hit the mark. The best result by analytic methods was obtained by Rankin [27] who showed that

$$\tau(n) = O(n^{\frac{11}{2} + \frac{3}{10}}).$$

Inspired by E. Artin, Weil [39] introduced in 1949 the zeta function of an algebraic variety over a finite field. He conjectured that this zeta function is a rational function which satisfies an appropriate functional equation and an analogue of the Riemann hypothesis. These formed the famous Weil conjectures. Shortly thereafter, experts in the theory of modular forms began to suspect a relationship between the Ramanujan conjecture (iii) and the Weil conjectures. Eichler [5], Ihara [10], Kuga and Shimura [12] began to search for a smooth projective variety X over $\mathbf{F}_p$ such that the polynomial

$$1 - \tau(p)T + p^{11}T^2$$

appears as a factor in the zeta function of X . They constructed an X which almost worked (the 10 fold product of a generic elliptic curve) but was not compact. Deligne [4] showed how to compactify their X and so the Ramanujan conjecture (iii) turned out to be a consequence of his proof of the Weil conjectures [3].

§2. Linnik - Selberg

Partly motivated by the Ramanujan conjecture (iii), Selberg [29] and Linnik [15] independently proposed the following conjecture. Let

$$S(m, n; c) = \sum_{\substack{a\bar{a} \equiv 1 \pmod{c} \\ (a, c) = 1}} e^{\frac{2\pi i}{c}(ma + n\bar{a})}$$

be the Kloosterman sum. Then:

Linnik - Selberg conjecture:

$$\sum_{c \leq x} \frac{S(1, n; c)}{c} \ll x^\epsilon.$$

where the implied constant depends only on ϵ .

Theorem 1. The Linnik-Selberg conjecture implies

$$\tau(p) = O_\epsilon(p^{\frac{11}{2} + \epsilon})$$

for every $\epsilon > 0$.

This fact was certainly known to Selberg [29, p. 9]. A proof appears in [18].

The set S_k of all cusp forms of weight k for $SL_2(\mathbf{Z})$ forms a finite dimensional vector space over $\mathbf{C}$. Let $k \geq 4$. An explicit basis for S_k can be constructed as follows. Define the m -th Poincaré series as

$$P_m(z) = \frac{1}{2} \sum_{(c,d)=1} \frac{e^{2\pi i m(\frac{az+b}{cz+d})}}{(cz+d)^k},$$

where the summation is over all pairs of integers c, d satisfying $(c, d) = 1$ and a, b are integers chosen so that $ad - bc = 1$. (The sum is independent of the choice, as can be easily checked.) $P_m(z)$ is a modular form of weight k for $SL_2(\mathbf{Z})$. If r is the dimension of S_k , then it was proved by Petersson [25] that $P_1, \dots, P_r$ form a basis of S_k . In particular, since S_{12} is one dimensional and generated by Δ , it follows that

$$P_1(z) = c\Delta(z),$$

for some non-zero constant c . The advantage in considering P_m is that one can derive an explicit formula for the n -th Fourier coefficient. An explicit calculation of the Fourier coefficient of P_1 reveals that

$$\tau(n) = 2\pi n^{11/2} \sum_{c=1}^{\infty} \frac{S(1, n; c)}{c} J_{11}\left(\frac{4\pi\sqrt{n}}{c}\right)$$

where J_{11} denotes the Bessel function of order 11. If J_{11} is replaced by J_{k-1} , then the resulting expression gives a formula for the n -th Fourier coefficient of a Poincaré series of weight k . Thus, if in the above sum J_{11} is replaced by J_9 , the resulting expression is identically zero as $S_{10} = \{0\}$.

Proof of Theorem 1. On the basis of the Selberg-Linnik conjecture it follows that

$$\sum_{c > \sqrt{n}} \frac{S(1, n; c)}{c} J_{11}\left(\frac{4\pi\sqrt{n}}{c}\right) \ll n^\epsilon$$

by partial summation and standard estimates for the Bessel function. As there are no cusp forms of weight 10, it also follows by a similar estimation that

$$\sum_{c < \sqrt{n}} \frac{S(1, n; c)}{c} J_9\left(\frac{4\pi\sqrt{n}}{c}\right) \ll n^\epsilon.$$

It remains to estimate

$$\sum_{c < \sqrt{n}} \frac{S(1, n; c)}{c} J_{11}\left(\frac{4\pi\sqrt{n}}{c}\right).$$

By utilising the key identity,

$$J_9(x) + J_{11}(x) = \frac{20J_{10}(x)}{x}$$

we are reduced to estimating

$$\frac{1}{\sqrt{n}} \sum_{c < \sqrt{n}} S(1, n; c) J_{10}\left(\frac{4\pi\sqrt{n}}{c}\right).$$

Familiar estimates for the Bessel function and a straightforward partial summation yields the desired result. This completes the proof.

The Ramanujan conjecture implies that

$$|\tau(n)| \leq n^{11/2} d(n)$$

where $d(n)$ denotes the number of divisors of n . From estimates on the divisor function, it follows that

$$\tau(n) = O(n^{11/2} \exp(\frac{c \log n}{\log \log n})),$$

for any positive constant $c > \log 2$. It was long conjectured that this was sharp in the following sense.

Theorem 2.

$$\tau(n) = \Omega_{\pm}(n^{11/2} \exp(\frac{c \log n}{\log \log n}))$$

for some constant $c > 0$.

Remarks.

1. We write $f(n) = \Omega_{\pm}(g(n))$ to mean that

$$f(n) > g(n)$$

for an infinity of n and

$$f(n) < -g(n)$$

for an infinity of n .

2. The constant c in this result is not optimal. On the Sato-Tate conjecture (discussed in §3) any $c > \log 2$ is valid.
3. This was proved in Ram Murty [19]. It has a long history. As a consequence of his conjectures (i), (ii), and (iii), Ramanujan [26] showed that

$$|\tau(n)| \geq n^{11/2}$$

for infinitely many values of n . Hardy [8] gave an unconditional proof of this. Then Rankin [28] proved that

$$\limsup_{n \rightarrow \infty} \frac{|\tau(n)|}{n^{11/2}} = +\infty.$$

Joris [11] improved upon this and showed that

$$\tau(n) = \Omega \left(n^{11/2} \exp \left(\frac{c(\log n)^{1/22}}{(\log \log n)^{23/22}} \right) \right).$$

Then, R. Balasubramanian and M. Ram Murty [2] proved that

$$\tau(n) = \Omega \left(n^{11/2} \exp((\log n)^{2/3-\epsilon}) \right).$$

The final result, which is best possible, was obtained in [19].

Theorem 2 can be used to prove:

Theorem 3.

$$\sum_{c \leq x} \frac{S(1, n; c)}{c} = \Omega \left(\exp \left(\frac{\log x}{\log \log x} \right) \right).$$

This says that we cannot expect too much cancellation to occur in the sum of Kloosterman sums that appears in the Linnik-Selberg conjecture. The idea of the proof is that if the sum in Theorem 3 is too small, then the explicit formula for $\tau(n)$ would imply that $\tau(n)$ itself is too small. This would then contradict Theorem 2. The details can be found in [19].

§3. Sato - Tate

If for each prime p , we write

$$\tau(p) = 2p^{11/2} \cos \theta_p$$

then it is conjectured that the θ_p 's are uniformly distributed in $[0, \pi]$ with respect to the measure

$$\frac{2}{\pi} \sin^2 \theta d\theta.$$

This was first proposed by Serre [32, p. I - 21] and is called the Sato - Tate conjecture because of the close analogy it bears to a conjecture of Sato and Tate on elliptic curves. Serre [32] suggested a general procedure for such questions. If for each prime p we associate a conjugacy class X_p in a compact group G , then given an irreducible representation ρ of G , define the L -series

$$L(s, \rho) = \prod_p \det(1 - \rho(X_p)p^{-s})^{-1}.$$

Serre [32] proved that if for every non-trivial irreducible representation ρ of G ,

- (a) $L(s, \rho)$ has an analytic continuation to $\operatorname{Re}(s) \geq 1$ and
- (b) $L(1 + it, \rho) \neq 0$ for $t \in \mathbf{R}$,

then the X_p 's are uniformly distributed in G with respect to the Haar measure of G .

In our case, let $G = SU_2(\mathbf{C})$. The conjugacy classes of G are parametrized by $0 \leq \theta \leq \pi$ and each class has a representative of the form

$$\begin{pmatrix} e^{i\theta} & \\ & e^{-i\theta} \end{pmatrix}.$$

The irreducible representations are given by the sequence $\rho_0, \rho_1, \dots$ where ρ_n has character χ_n given by

$$\chi_n \begin{pmatrix} e^{i\theta} & \\ & e^{-i\theta} \end{pmatrix} = \frac{\sin(n+1)\theta}{\sin \theta}.$$

Therefore, Serre's theorem states that if for $n \geq 1$,

- (a) $L(s, \rho_n)$ has an analytic continuation to $\operatorname{Re}(s) \geq 1$ and
- (b) $L(1 + it, \rho_n) \neq 0$, for $t \in \mathbf{R}$,

then the Sato-Tate conjecture is true. Kumar Murty [23] subsequently showed that

(a) alone suffices to imply the Sato-Tate conjecture. That is, the non-vanishing turns out to be a consequence of the analytic continuation to the line $\operatorname{Re}(s) = 1$.

Not much is known about $L(s, \rho_n)$. For instance, if $n = 0$, then $L(s, \rho_0) = \zeta(s)$, the Riemann zeta function. If $n = 1$, then

$$L(s, \rho_1) = \sum_{n=1}^{\infty} \frac{\tau(n)}{n^{s+\frac{11}{2}}}$$

is the series introduced by Ramanujan [26] and further studied by Hecke [9]. For $n = 2$, Rankin [27] and Selberg [30] independently showed that $\zeta(s)L(s, \rho_2)$ has an analytic continuation and a functional equation, except for a simple pole at $s = 1$. Shimura [36] finally showed that $L(s, \rho_2)$ is in fact entire. Recently, Shahidi [35] proved that $L(s, \rho_3), L(s, \rho_4), L(s, \rho_5)$ each has an analytic continuation for $\operatorname{Re}(s) \geq 1$.

Let $\pi(x)$ denote the number of primes up to x . The Sato - Tate conjecture is equivalent to the assertion that for each k ,

$$(A_k) \quad \sum_{p \leq x} (2 \cos \theta_p)^{2k} \sim \frac{1}{k+1} \binom{2k}{k} \pi(x)$$

as $x \rightarrow \infty$. One can prove [19] :

Theorem 4. If for $n \leq N$, $L(s, \rho_n)$ has analytic continuation for $\operatorname{Re}(s) \geq 1$, then A_k is true for $k \leq [N/2]$.

In particular, Theorem 4 is true for $N = 4$ and hence,

$$\sum_{p \leq x} (2 \cos \theta_p)^4 \sim 2\pi(x)$$

as $x \rightarrow \infty$.

Proof of Theorem 2. It follows from (A_2) that for some $\delta > 0$, there are at least $\delta x / \log x$ primes satisfying

$$|2 \cos \theta_p| > 2^{1/4}, \quad \frac{x}{2} < p < x.$$

Let n be defined by

$$\log n = \sum_{\substack{\frac{x}{2} < p < x \\ |2 \cos \theta_p| > 2^{1/4}}} \log p.$$

Then,

$$|\tau(n)| \geq n^{11/2} \exp\left(\frac{c\delta x}{\log x}\right)$$

where $c = \frac{\log 2}{4}$. Since, $\delta x \leq \log n \leq 2x$, it follows that

$$|\tau(n)| \geq n^{11/2} \exp\left(\frac{c_1 \log n}{\log \log n}\right)$$

for some $c_1 > 0$. This completes the proof.

§4. Lehmer

Based on empirical evidence, Lehmer [14] conjectured that $\tau(p) \neq 0$, for every prime p . This is equivalent to the non-vanishing of the Poincaré series P_n of weight 12, for every natural number n . This is unknown at present.

In 1967, Serre [33] conjectured the existence of an ℓ -adic representation

$$\rho_\ell : \text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \rightarrow GL_2(\mathbf{Z}_\ell)$$

such that if σ_p is the Frobenius automorphism of p , then

$$\text{tr } \rho_\ell(\sigma_p) = \tau(p), \quad \det \rho_\ell(\sigma_p) = p^{11}.$$

This conjecture was proved by Deligne [4]. Serre [34] noticed that this fact can now be utilised, in conjunction with the Chebotarev density theorem, to show that $\tau(p) = 0$ very rarely. If we consider reduction (mod ℓ),

$$GL_2(\mathbf{Z}_\ell) \rightarrow GL_2(\mathbf{F}_\ell)$$

then an extension $K_\ell/\mathbf{Q}$ is determined by the kernel of the composition of this map with ρ_ℓ . This extension has the property that if σ_p represents the Frobenius automorphism of $\text{Gal}(K_\ell/\mathbf{Q})$ then

$$\begin{aligned} \text{tr } \rho_\ell(\sigma_p) &\equiv \tau(p) \pmod{\ell} \\ \det \rho_\ell(\sigma_p) &\equiv p^{11} \pmod{\ell} \end{aligned}.$$

Serre and Swinnerton-Dyer [38] showed that for sufficiently large ℓ ,

$$\text{Gal}(K_\ell/\mathbf{Q}) = \{g \in GL_2(\mathbf{F}_\ell) : \det g \in (\mathbf{F}_\ell^*)^{11}\}.$$

The number of primes $p \leq x$ such that $\tau(p) = 0$ is contained in the set of primes such that $\tau(p) \equiv 0 \pmod{\ell}$. A quick calculation of the number of elements of trace zero

in $\text{Gal}(K_\ell/\mathbf{Q})$ and an application of the effective Chebotarev density theorem shows that

$$\#\{p \leq x : \tau(p) = 0\} \ll \frac{x}{(\log x)^{1+\gamma}}$$

for some $\gamma > 0$. This was proved by Serre [34]. Serre [34] further showed that if the generalised Riemann hypothesis for Artin L -series is assumed then

$$\#\{p \leq x : \tau(p) = 0\} = O(x^{3/4}).$$

§5. Lang-Trotter

As part of more general conjectures on Fourier coefficients of cusp forms, Lang and Trotter [13] have conjectured that if for each integer a , we set

$$\pi_a(x) = \#\{p \leq x : \tau(p) = a\}$$

then $\pi_a(x) = O_a(1)$. That is, for each a , the equation $\tau(p) = a$ has only finitely many solutions. A variant of this conjecture was proved recently by M. Ram Murty, V. Kumar Murty and T. N. Shorey [17] for odd values of a . More precisely:

Theorem 5. If a is odd, and $\tau(n) = a$ then $n \leq \exp((2|a|)^c)$ for some effectively computable constant $c > 0$.

Corollary. If $\tau(n)$ is odd, then $|\tau(n)| \geq (\log n)^c$ for some effectively computable absolute constant $c > 0$.

The starting point is the observation that $\tau(p)$ is even for every prime p . To see this note that

$$\begin{aligned} \sum_{n=1}^{\infty} \tau(n)q^n &= q \prod_{n=1}^{\infty} (1 - q^n)^{24} \\ &\equiv q \prod_{n=1}^{\infty} (1 + q^{8n})^3 \pmod{2} \\ &\equiv q \sum_{n=0}^{\infty} q^{4n^2+4n} \pmod{2} \end{aligned}$$

by an identity of Jacobi. Thus, $\tau(n)$ is odd if and only if $n = (2m+1)^2$.

A natural number n is called *squarefull* if for every prime $p|n$, we have $p^2|n$. Thus, if $\tau(n)$ is odd, then n is squarefull. By using techniques from transcendence,

it is possible to prove that there is an effectively computable positive constant c such that

$$|\tau(n)| \geq (\log n)^c$$

whenever n is squarefull. Following Ramanujan [26] we can write

$$\tau(p^m) = \frac{\alpha_p^{m+1} - \bar{\alpha}_p^{m+1}}{\alpha_p - \bar{\alpha}_p}$$

utilising the truth of (ii) and (iii). If $\tau(p) \neq 0$, it follows that

$$\begin{aligned} |\tau(p^m)/\tau(p)| &= \left| \frac{\alpha_p^{m+1} - \bar{\alpha}_p^{m+1}}{\alpha_p^2 - \bar{\alpha}_p^2} \right| \\ &\geq \frac{1}{2} p^{\frac{11}{2}(m-1)} \left| \left(\frac{\alpha_p}{\bar{\alpha}_p} \right)^{m+1} - 1 \right|. \end{aligned}$$

There is a theorem of Baker [1] which states that if α is an algebraic number, then there is an effectively computable positive constant c such that $\alpha^n = 1$ or

$$|\alpha^n - 1| > \exp(-cH(\alpha) \log n)$$

where $H(\alpha)$ denotes the logarithm of the absolute values of the coefficients of the minimal polynomial of α over $\mathbf{Q}$. If we apply this here, we obtain

$$|\tau(p^m)| \geq |\tau(p)| p^{\frac{11}{2}(m - c_1 \log m)}$$

for some $c_1 > 0$. This gives us a good lower bound provided m is sufficiently large, say for $m \geq C_2$.

For small values of m , a subtler approach is needed. The key observation is that

$$\tau(p^m) = \gamma_m(p) \prod_{r=1}^{\lfloor m/2 \rfloor} (\tau(p)^2 - 4p^{11} \cos^2 \frac{\pi r}{m+1})$$

where

$$\gamma_m(p) = \begin{cases} 1 & \text{if } m \text{ is even,} \\ \tau(p) & \text{if } m \text{ is odd.} \end{cases}$$

That is, $\tau(p^m)$ is a binary form in $\tau(p)^2$ and p^{11} . Moreover, this binary form has degree ≥ 3 when $m \geq 6$. There is a result of Feldman [6] which states that if f is

an irreducible binary form of degree ≥ 3 with rational integer coefficients and $m \neq 0$, then the equation $f(x, y) = m$ forces

$$\max\{|x|, |y|\} \leq m^C$$

for some effectively computable constant C depending only on f . Applying this to the case of the $\tau(p^m)/\gamma_m(p)$ function, we obtain for $m \geq 6$ that

$$|\tau(p^m)| \geq p^C$$

where C depends only on m . Thus, for $6 \leq m \leq C_2$, there is an absolute constant $c_3 > 0$ such that

$$|\tau(p^m)| \geq p^{mc_3}.$$

Combining this with our previous lower bound enables us to deduce that

$$|\tau(p^m)| \geq p^{mc}$$

for some effectively computable positive constant c . To treat $2 \leq m \leq 5$, the approach is similar, but the results are not as sharp. For instance,

$$\tau(p)^2 = p^{11} + \tau(p^2)$$

so that the point $(p, \tau(p))$ lies on the hyperelliptic curve

$$y^2 = x^{11} + a,$$

with $a = \tau(p^2)$. A theorem of Sprindzuk [37] on the solutions of hyperelliptic equations tells us that if (x, y) lies on the above curve, then,

$$\log \max\{|x|, |y|\} \ll |a|^C$$

for some effectively computable positive constant C . This implies that

$$|\tau(p^2)| \geq (\log p)^c$$

for some positive constant c . Putting all these facts together, yields the result that if n is squarefull then, either $\tau(n) = 0$ or

$$|\tau(n)| \geq (\log n)^C$$

for some positive constant C .

Serre [34] showed that if we assume the generalised Riemann hypothesis, then

$$\pi_a(x) = O(x^{7/8}).$$

This was improved in [22] to:

Theorem 6. Assuming *GRH*, for $a \neq 0$,

$$\pi_a(x) = O(x^{4/5}).$$

The proof of Theorem 6 is based on ℓ -adic representations and an improved version of the Chebotarev density theorem. This improvement is based on a simple group theoretic idea. Let G be a finite group and let

$$\pi : \hat{G} \rightarrow \mathbf{C}$$

be a function defined on the irreducible characters of G . For each conjugacy class C of G , let g_C be a representative of C and define

$$\pi_C = \frac{|C|}{|G|} \sum_{\chi} \overline{\chi(g_C)} \pi(\chi)$$

where the summation is over irreducible characters of G . The orthogonality relations imply that

$$(\dagger) \quad \sum_C \frac{1}{|C|} |\pi_C - \frac{|C|}{|G|} \pi(1)|^2 = \frac{1}{|G|} \sum_{\chi \neq 1} |\pi(\chi)|^2$$

where the first sum is over conjugacy classes C of G .

Now let $K/\mathbf{Q}$ be a normal extension with Galois group G . For each prime p , let σ_p denote the Frobenius automorphism and set

$$\pi(x, \chi) = \sum_{p \leq x} \chi(\sigma_p).$$

Assuming Artin's conjecture on the holomorphy of $L(s, \chi)$ and the analogue of the Riemann hypothesis, it is essentially deduced that for $\chi \neq 1$,

$$\pi(x, \chi) = O(\chi(1)x^{1/2} \log x),$$

apart from a few minor logarithmic factors. Now,

$$\pi_C(x) = \frac{|C|}{|G|} \sum_{\chi} \overline{\chi(g_c)} \pi(x, \chi)$$

enumerates the number of primes $p \leq x$ such that $\sigma_p \in C$. Therefore, the above identity implies that

$$\frac{1}{|C|} |\pi_C(x) - \frac{|C|}{|G|} \pi(x)|^2 = O(x \log^2 x)$$

in view of the group theoretic identity

$$\sum_{\chi} \chi(1)^2 = |G|.$$

Thus, ignoring a few logarithmic factors,

$$(\dagger) \quad |\pi_C(x) - \frac{|C|}{|G|} \pi(x)| \ll |C|^{1/2} x^{1/2} \log x.$$

By using the Cauchy-Schwarz inequality and the identity $(\dagger)$, it follows that the same estimate is true for any set C stable under conjugation.

The number of primes $p \leq x$ such that $\tau(p) = a$ is certainly bounded by the number of primes $p \leq x$ satisfying

$$(*) \quad \tau(p) \equiv a \pmod{\ell}.$$

By the result of Serre and Swinnerton-Dyer [38] on the shape of the Galois group, cited earlier, the primes p satisfying the above congruence have Frobenius automorphism $\sigma_p \in GL_2(\mathbf{F}_{\ell})$ of trace a . This comprises a set C_{ℓ} which is stable under conjugation. Moreover, the size of this set is roughly ℓ^3 , so that the number primes satisfying $(*)$ is

$$\ll \frac{1}{\ell} \pi(x) + O(\ell^{3/2} x^{1/2}).$$

Choosing $\ell = x^{1/5}$ yield the result

$$\pi_a(x) = O(x^{4/5})$$

which is Theorem 6. It would appear that we have assumed the truth of Artin's conjecture. This hypothesis can be removed by very nice arguments of a group-theoretic nature. The details can be found in [22].

If $a = 0$, Serre [34] proved that $\pi_0(x) = O(x^{3/4})$. Kumar Murty [24], proceeding differently, proved the following:

Theorem 7. If $L(s, \rho_n)$ have an analytic continuation to the entire complex plane, satisfy appropriate functional equations for $n \geq 1$ and the analogue of the Riemann hypothesis, then

$$\pi_a(x) = O(x^{3/4})$$

for all integers a .

It should be noted that the ℓ -adic approach does not yield as good a result for all a .

§6. Atkin-Serre

There is an even more ambitious conjecture about the growth of $\tau(p)$.

Namely, Atkin-Serre [31, eqn. 4.11k] have conjectured that

$$|\tau(p)| \gg_{\epsilon} p^{9/2-\epsilon}.$$

Using methods of analytic number theory, we can obtain information for almost all primes p .

For instance, let $\nu(n)$ denote the number of prime factors of n . It is a classical result of Hardy and Ramanujan [7] that

$$\sum_{n \leq x} (\nu(n) - \log \log n)^2 = O(x \log \log x).$$

Such a result implies that

$$(\star) \quad |\nu(n) - \log \log n| < (\log \log n)^{\frac{1}{2}+\epsilon}$$

for all $n \leq x$ apart from

$$O\left(\frac{x}{(\log \log x)^{\epsilon}}\right)$$

exceptional values of n . That is, a natural number “usually” has $\log \log n$ prime factors. If n satisfies $(\star)$, it is called a normal number. If we treat $\tau(p)$ as a normal number, we would expect that it too should have “usually” $\log \log p$ prime factors. This is indeed the case. One can establish the following “modular analogue” :

Theorem 8. On the generalised Riemann hypothesis for Dedekind zeta functions (*GRH*)

$$\sum_{\substack{p \leq x \\ \tau(p) \neq 0}} (\nu(\tau(p)) - \log \log p)^2 = O\left(\frac{x \log \log x}{\log x}\right).$$

This implies that for $p \leq x$,

$$|\nu(\tau(p)) - \log \log p| < (\log \log p)^{\frac{1}{2} + \epsilon}$$

apart from

$$O\left(\frac{x}{(\log x)(\log \log x)^\epsilon}\right)$$

exceptional primes. If we now consider $\tau(n)$, then it seems plausible to expect

$$\sum_{p|n} \log \log p$$

prime divisors in view of the fact that $\tau(n)$ is multiplicative. The above sum behaves like

$$\frac{1}{2}(\log \log n)^2$$

on the average. This heuristic reasoning is also correct and in a similar way, one can prove [20]: on GRH,

$$\sum_{\substack{n \leq x \\ \tau(n) \neq 0}} (\nu(\tau(n)) - \frac{1}{2}(\log \log n)^2)^2 = O(x(\log \log x)^{3+\epsilon})$$

More generally:

$$Pr\left\{\frac{\nu(\tau(n)) - \frac{1}{2}(\log \log n)^2}{(\log \log n)^{3/2}} \leq \frac{\alpha}{\sqrt{3}}\right\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\alpha} e^{-t^2/2} dt.$$

This is the “modular analogue” of the celebrated Erdős - Kac theorem.

The implications of these results is that if $\tau(p)$ has usually $\log \log p$ distinct prime factors, then it cannot be too small. The full strength of the GRH is not necessary. Let

$$\pi(x, \ell) = \#\{p \leq x : \tau(p) \equiv 0 \pmod{\ell}\}.$$

Then the Chebotarev density theorem implies that

$$\pi(x, \ell) \sim \delta(\ell) li \ x$$

where

$$li\ x = \int_2^x \frac{dt}{\log t}.$$

We make the hypothesis:

$$(H) \quad \sum_{\ell \leq X} |\pi(x, \ell) - \delta(\ell) li\ x| \ll \frac{x}{(\log x)^3}$$

for $X = \exp(\log x / \log \log x)$. (H) is a consequence of GRH but is considerably weaker. It can be viewed as a non-abelian analogue of the Bombieri - Vinogradov theorem on primes in arithmetic progressions.

The proof of Theorem 8 requires asymptotic formulas for

$$\sum_{\substack{p \leq x \\ \tau(p) \neq 0}} \nu(\tau(p)), \quad \sum_{\substack{p \leq x \\ \tau(p) \neq 0}} \nu^2(\tau(p)).$$

In order to see how this is done, let us consider the first sum:

$$\begin{aligned} \sum_{\substack{p \leq x \\ \tau(p) \neq 0}} \nu(\tau(p)) &= \sum_{\substack{p \leq x \\ \tau(p) \neq 0}} \left(\sum_{\substack{\ell | \tau(p) \\ \ell < X}} 1 + \sum_{\substack{\ell | \tau(p) \\ \ell > X}} 1 \right) \\ &= \sum_{\ell < X} \pi(x, \ell) + O\left(\frac{x \log \log x}{\log x}\right). \end{aligned}$$

The desired asymptotic formula follows from (H) on noting that

$$\delta(\ell) \sim 1/\ell.$$

The second sum is handled similarly.

Results of this nature allow us to obtain lower bounds for $\tau(p)$ for almost all p .

Theorem 9. For all but $O(\pi(x)(\log \log x)^{-\epsilon})$ primes $p \leq x$,

(i) $|\tau(p)| \geq (\log p)^{1-\epsilon}$, unconditionally,

(ii)

$$|\tau(p)| \geq \exp((\log p)^{1-\epsilon}),$$

assuming (H).

The improved effective Chebotarev density theorem [22] leads to the following:

Theorem 10. On GRH,

$$|\tau(p)| > p^{\frac{1}{4}-\epsilon}$$

with at most $O(x^{1-\epsilon})$ exceptional primes $p \leq x$.

The proof of Theorem 10 is curious. In view of Theorem 6, it is only immediate that

$$|\tau(p)| \geq p^{\frac{1}{5}-\epsilon}$$

for all but $O(x^{1-\epsilon})$ primes $p \leq x$. The improvement is obtained by considering the union of all conjugacy classes of trace $< X^{\frac{1}{4}-\epsilon}$ and then invoking the estimate (§).

These results reveal to some extent the arithmetic significance of the τ function and Fourier coefficients of cusp forms in general. The story is not yet over. The future awaits for some deeper connections between non-abelian class field theory and Fourier coefficients of cusp forms.

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